

EXECUTIVE SUMMARY

AI Agents at the Border

The Executive Decision Brief

VALUE · AUTONOMY · SOVEREIGNTY · EXECUTION

THE OPERATING PRINCIPLE

Decisions may transfer within a signed mandate.
Accountability does not.

REGULATORY CUT-OFF · 10 JULY 2026

Re-verify applicability and deadlines before procurement or any autonomy increase.

MAKSYM NECHEPURENKO

Dubai · 2026 · v1.9.2

A Decision Brief, Not a Substitute

This edition is the fast route through *AI Agents at the Border*. It is written for an executive team that needs a shared decision language, a bounded first portfolio, and a ninety-day operating plan before it needs the full evidential and implementation detail.

The brief preserves the book's central discipline: decisions may transfer within a signed mandate; accountability does not. It also preserves the evidence labels. A customs outcome, an adjacent-domain method, a consultancy-reported case, and a directional hypothesis are not interchangeable merely because they fit on the same slide.

This brief now carries the minimum working set needed to begin: a decision sequence, mandate card, RFP decision-rights clause, regulatory applicability checklist, pilot gate, and board dashboard. A leadership team can use it to define the first task, commission a controlled procurement, complete an initial legal screen, and make a pilot decision. The full book remains the reference for expanded clauses, jurisdiction detail, evidence, metrics, and implementation depth - especially Chapter 6 for readiness, Chapter 7 for procurement, Chapters 8-10 for legal, assurance, and security design, and Chapter 12 for pilot execution - not a prerequisite for starting.

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How to Use This Brief

This brief is for a leadership team that has to turn *we should do something with AI* into a governed portfolio. It does not reproduce every clause, metric, case, and source from the reference edition. It does contain the minimum tools required to define a task, establish decision rights, screen legal applicability, commission a pilot, and make a gate decision.

Read it once as a team. On the second pass, complete the gates in Chapter 1 in sequence, assign an owner to every action in Chapter 8, and use the Core Tools A-E. The full book adds implementation depth and jurisdiction-specific research; it is no longer required to begin the work. For depth, use Chapter 5 for value economics, Chapter 6 for readiness, Chapter 7 for procurement and vendor control, Chapters 8–9 for legal and assurance design, Chapter 10 for agentic security, and Chapter 12 for pilot execution.

IMPORTANT — REGULATORY CUT-OFF · 10 JULY 2026. Law, deadlines, guidance, and standards change. Before procurement, production approval, or any autonomy increase, local counsel must re-verify task applicability and dates. Re-check Sections 8.2–8.4 and [the regulatory watch record](#). This is a dated baseline, not legal advice, certification, or a compliance guarantee.

The Stance

MOTIF · CH 1 / CH 16

Three commitments, held together

The book opens and closes on these three. They only work as a set.



The Stance

The stance in one page

The book holds three commitments simultaneously.

Ambitious on value. The target is not a decorative chatbot or a ten percent improvement in a step that should not exist. It is a changed cost of control: more trade and more adaptive fraud handled without matching growth in officer effort, while facilitation becomes faster and more predictable.

Conservative on autonomy. A system earns authority one task and one rung at a time. A fluent model is not evidence that it may act. Promotion requires a local record of quality, exceptions, cost, security, human capacity, and reversibility.

Uncompromising on sovereignty and trust. Trade data, public authority, and the evidence needed to audit both remain under lawful administrative control. Consumer subscriptions, personal API keys, and unreviewed data paths are excluded territory. A commercial cloud region is not a sovereign deployment merely because it is nearby.

Four evidence labels

- **Verified:** a named administration or primary operating source identifies the technology and outcome.
- **Adjacent:** a neighboring domain supports a method or failure mode, not a customs forecast.
- **Practice-reported:** a public consultancy or delivery case; useful, attributed, and not independently verified.
- **Directional:** a hypothesis to test locally, not an observed result.

The label travels with the number. If it disappears when a claim enters a ministerial slide, the claim should disappear too.

INDEPENDENT WORK. *The author writes in a personal capacity. This is not an Agility publication and does not represent or imply approval or endorsement by the author’s employer, its clients, the WCO, or any cited administration. All cases and reported outcomes come from attributable public sources; none describes an Agility or client engagement.*

The executive starter kit

Use the five Core Tools as one operating sequence. Each produces the input required by the next; none is a standalone compliance certificate.

Tool	Decision produced
A. Mandate card	Task, constraint, authority ceiling, sovereign path, owner, fallback, and validity.
B. Minimum metric spine	Baseline, target, stop threshold, evidence, owner, and action across the operating record.
C. Applicability checklist	Legal role, affected people, risk class, binding duties, controls, evidence, and open counsel questions.
D. RFP decision-rights clause	Retained authority, supplier limits, evidence, material change, acceptance, and exit.
E. Pilot gate	Promote, hold, demote, or stop against pre-committed evidence and a residual-risk owner.

The 90-minute working session

Time	Work	Product
0-10	Name the operating constraint and owner.	Gate 1 statement
10-25	Complete Core Tool A; resolve or assign Gates 2-4 gaps.	Draft mandate card
25-40	Complete Core Tool B against the current baseline.	Metric owners, targets, stop thresholds
40-55	Complete Core Tool C using official sources.	Applicability record and counsel questions
55-70	Adapt Core Tool D to the task and procurement route.	Decision-rights schedule
70-90	Set the first evidence gate with Core Tool E.	Pilot stage, gate date, signers

Minimum working-session output: one selected task · completed Gates 1-4 or named owners for every gap · draft mandate card · metric owners · legal screen · procurement decision-rights schedule · pilot gate date.

If the session starts with a vendor or model, return to Gate 1. If a later tool changes the task, authority, data path, or jurisdiction, re-open the earliest affected gate rather than patching the paperwork downstream.

The Eight Decisions

Start with the mandate, not the model

Customs leaders are not choosing whether AI exists. They are deciding where software may enter a public-authority workflow, what it may do there, which evidence earns more authority, and who remains answerable when it is wrong. Those are operating decisions. Model selection comes later.

DECIDE	EVIDENCE	STOP
Select one operating constraint and a bounded first task.	Named owner, measured baseline, authority ceiling, and sovereign path.	No value case, RFP, or pilot while any foundation gate remains open.

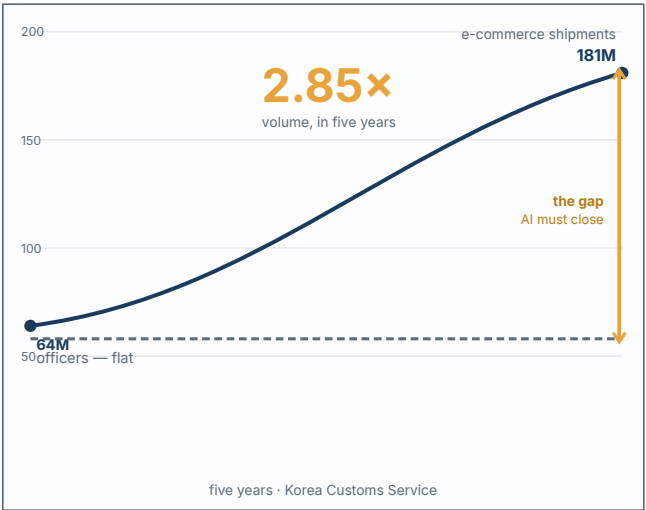
The arithmetic is already familiar: parcel and declaration volumes rise; fraud industrialises and adapts; experienced officers remain scarce; traders expect a digital service; government expects control and revenue without linear headcount. Classical rules and machine

The Arithmetic

CHAPTER 1

Why the mandate now needs more than headcount

Transaction volume grows far faster than any workforce plan. The widening gap is the case for AI.



Verified · administration-stated (KCS, WCO News 108).

Volume outgrows any workforce plan — the gap is the case for AI.

Figure 1.1 — The Arithmetic

learning remain essential, but generative and agentic systems add a new option: software can read the unstructured record, assemble evidence, choose a next action, and carry work across systems. That last capability changes the governance question from *is the answer accurate?* to *what authority did the system exercise?*

The foundation rule

The executive programme can be reduced to eight decisions, but they are not eight menu items. They form a dependency chain. **Do them in order. Each gate must produce its minimum evidence before the next gate opens.** The first four gates establish whether there is a governable task. The last four decide whether and how to commit resources. Starting with a vendor, model, or promised saving skips the object that procurement and assurance need to govern.

The first four are prerequisites, not parallel workstreams:

1. **Constraint:** an observable operating problem and accountable owner.
2. **Task and baseline:** a bounded workflow measured before intervention.
3. **Autonomy and agency:** a signed decision and permission ceiling, including actions that remain technically impossible.
4. **Sovereignty:** a lawful data/deployment path with evidence custody, fallback, and exit.

BOX 1.A Foundation stop rule

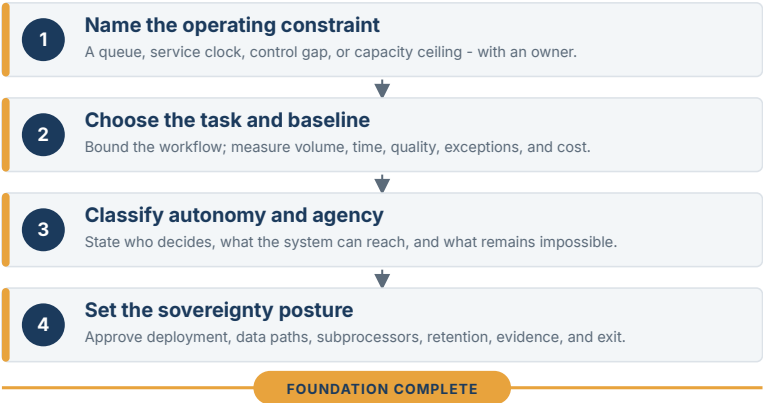
Do not approve a value case, sourcing route, RFP, pilot, or autonomy increase while any foundation gate is open. A gap may have a named owner and resolution date; it does not become implied permission for a later gate.

The Decision Dependency Chain

EXECUTIVE SUMMARY

Do these in order. Evidence opens the next gate - confidence does not.

FOUNDATION GATES



Gate	Completion condition before the next gate
2. Task and baseline	Workflow boundary plus volume, time, quality, exceptions, unit cost, and owner.
3. Autonomy and agency	Decision rung plus tools, data, identities, permissions, writes, limits, and impossible actions.
4. Sovereignty posture	Lawful deployment, data flows, subprocessors, retention, evidence custody, and exit boundary.
5. Value bridge	Output-to-action conversion, value range, permanent human/assurance cost, and benefit owner.
6. Ownership route	Build-buy-partner boundary, retained capability, commercial comparison, and exit route.
7. Right to operate	Signed mandate, thresholds, human route, security tests, operational assurance record, fallback, and demotion trigger.
8. Renewal rhythm	Review cadence, evidence owner, next decision date, and continue-change-compete-demote-retire options.

Dependency rule. Gates 5-8 are invalid while any of Gates 1-4 remains open. If the constraint, baseline, authority, or sovereign path changes, reopen the chain at the earliest affected gate.

The first portfolio is deliberately uneven

An administration does not become “AI-ready” as a single event. It may be ready now for read-only document extraction, nearly ready for a grounded procedure assistant, and unready for a write-capable

agent affecting a trader. That is normal. A useful portfolio contains three kinds of work:

- **value now:** bounded L1/L2 services using approved data and human confirmation;
- **foundations:** corpus stewardship, data contracts, identity, logging, evaluation, procurement, and supervisor capability;
- **options:** higher-autonomy tasks held in shadow mode until evidence and control capacity exist.

This avoids two symmetrical errors. The first is paralysis: treating every AI use as if it were an autonomous enforcement decision. The second is frontier theatre: funding an impressive agent before the administration can supply lawful data, absorb exceptions, observe trajectories, or stop the workflow safely.

The leadership test

Ask the proposed executive owner to complete one sentence:

For [named task], the administration will allow [system] to operate at [autonomy rung] with [agency profile], because [local evidence] meets [pre-committed thresholds]; [named person] remains accountable, and the task will be demoted or stopped when [trigger] occurs.

If the sentence cannot be completed, the project is not ready for a model discussion. If it can, technology, legal, procurement, security, and operations finally have the same object to design.

Core Tool A: Mandate card

Complete one card per task and proposed rung. If a field is unknown, the card records an owner and date for resolving it; it does not convert the gap into implied permission.

Mandate field	Decision record
Task and owner	Workflow: ____ · accountable owner: ____ · operating owner: ____
Constraint and baseline	Constraint: ____ · volume/time/quality/exceptions/unit cost: ____
Outcome and value	Target: ____ · conversion action: ____ · benefit owner: ____
Authority ceiling	Rung: ____ · agency: ____ · retained decisions/exclusions: ____
Sovereign path	Deployment/region: ____ · data/tools/identity: ____ · retention/evidence custodian: ____
Evidence to operate	Thresholds: ____ · human capacity: ____ · security/assurance record: ____
Containment and validity	Fallback/stop trigger: ____ · incident owner: ____ · volume/duration/renewal: ____

Approval record

Approval function	Name	Signature	Date
Operations	_____	_____	_____
Legal / privacy	_____	_____	_____
Security	_____	_____	_____
Data / IT	_____	_____	_____
Accountable executive	_____	_____	_____

Agent, Autonomy, and Agency

2

The definition that belongs in the RFP

An AI agent is a software system that pursues an assigned goal by planning multi-step work and using tools—systems, databases, and documents—with a defined degree of autonomy and under defined human oversight.

DECIDE	EVIDENCE	STOP
Set the autonomy rung and the separate agency profile for one task.	Decision rights, permitted tools/data/writes, limits, owner, and expiry.	A vendor cannot separate who decides from what the system can touch.

Each phrase is a control. The goal bounds purpose. Planning creates the possibility of adaptive behaviour. Tools define reach. Autonomy allocates decision rights. Oversight retains human command. Remove any one and the buyer no longer knows what is being procured.

The bright-line test: can one decision-making component choose the next permitted action; can that choice change because of an intermediate result; and can it stop, re-plan, or escalate instead of following a fixed route? A pre-written chain remains a workflow or pipeline even when several steps use AI. Adaptive selection from bounded actions is agentic. This is the practical defence against agent washing.

Rules, machine learning, generative AI, and agents are not replacement generations. A robust customs workflow stacks them: rules enforce hard boundaries, ML supplies risk signals, GenAI reads and drafts, and an agent may orchestrate the allowed sequence. A vendor that proposes to replace hard constraints with a prompt has confused fluent language with control.

The Autonomy Ladder

Four autonomy levels — pair every rung with a separate agency profile

Autonomy rises left to right: operational decisions pass from the officer to the agent — accountability does not.



Human — customs officer AI agent Amber underline = decision authority

Level colour deepens with autonomy; every distinction also holds in greyscale (shape + lightness).

Figure 2.1 — The Autonomy Ladder AI Agents at the Border · Chapter 2

Figure 2.1 — The Autonomy Ladder

L1 Assistant — Operator. The system drafts, retrieves, extracts, or summarises. A human checks every output before use. This is the default entry point for internal knowledge and document work.

L2 Copilot — Pilot. The system proposes an action inside the workflow; the officer confirms each consequential action. HS-code suggestion and document-completeness recommendations usually belong here.

L3 Supervised Agent — Supervisor. The system completes a bounded task across tools, while a human handles exceptions and retains the consequential decision. Case-file assembly is a strong candidate when the write path remains narrow and reversible.

L4 Autonomous Agent — Governor. The system decides and acts inside a machine-checkable mandate; humans audit outcomes and anomalies. It is defensible only for low-consequence, reversible, tightly bounded work with complete records and a tested fallback. Penalties and other materially adverse public-authority decisions do not earn this rung merely because a model performs well.

Promotion occurs one task and one rung at a time. A platform has no single autonomy level. On the same architecture, a search assistant may be L1, a classification suggestion L2, and a case-preparation loop L3.

Agency is the second dial

Autonomy asks who decides whether an action should occur. Agency asks what the software is technically able to touch: tools, data, permissions, write actions, communication channels, and transaction limits. The two must be recorded separately.

A read-only agent may adapt its research autonomously yet create little direct operational exposure. A copilot may require confirmation but still hold dangerous broad credentials. The control rule is **least agency**: grant only the specific capability needed for the task, use a distinct machine identity, deny everything else, and remove unused reach after observation.

A usable profile records six lines: **data scope** (document/case/queue/estate) · **named tools** · **write scope** (draft/staged/committed) · **communications** · **case, value, time, rate, and geography limits** · **distinct identity, owner, expiry, and revocation**.

BOX 2.A Two customs profiles that reverse the intuition

L2 + broad agency - dangerous. An HS copilot proposes and an officer confirms, yet its identity can read declarations, trader, inspection, and intelligence records; write notes; and send messages. Confirmation controls the classification, not exposure, unintended writes, or credential misuse.

L3 + narrow agency - manageable. A supervised agent chooses extraction and reconciliation steps inside one declaration, reads approved documents, and writes a draft checklist. It cannot search other cases, message the trader, change a risk score, or commit core data; the officer handles exceptions and the consequential decision.

Conclusion: L2 with broad credentials can expose more than L3 with a one-case, allowlisted, read-only profile. Record both dials.

The mandate card

Use Core Tool A; no completed card means no authority to operate. Before approval record: task · rung and why not lower · agency and why every permission is needed · mandate and signer · reconstructable evidence · oversight and demotion. A change to behaviour, evidence, data, tools, or authority re-enters the gate; a vendor-deck answer is not a control.

Value, Portfolio, and Readiness

3

The ladder of institutional readiness

The fastest credible route to agents begins below agents.

DECIDE	EVIDENCE	STOP
Choose the value channel and the task-specific readiness ceiling.	Local unit cost, conversion owner, governed data, and human capacity.	A benefit percentage arrives without a local baseline or conversion action.

The Maturity Ladder

Three stages of AI capability in customs

Value and capability rise stage by stage — but each rung depends on the prerequisites of the one below it.



Tones borrow from the Autonomy Ladder (Ch. 2); this ladder has three rungs, not four. Distinctions hold in greyscale.

Figure 3.1 — The Maturity Ladder

Document intelligence makes invoices, certificates, declarations, rulings, and correspondence machine-readable. **Grounded assistants** use approved corpora and return citations, corrections, and explicit uncertainty. **AI agents** then use those governed tools and evidence to close bounded loops. Skipping the lower rungs does not create speed; it pushes document, data, and verification defects into a system now able to act on them.

Start with a portfolio, not a single flagship. Score tasks on value, frequency, data readiness, reversibility, legal consequence, integration, human exception capacity, and security exposure. Set an autonomy ceiling before a vendor proposes one. The first agent should watch a defined queue, gather authorised evidence, prepare a cited recommendation, request a human decision, and write back only the approved result. Its most valuable early output is often the operational record: missing data, recurring exceptions, override reasons, unit cost, and useful tool permissions.

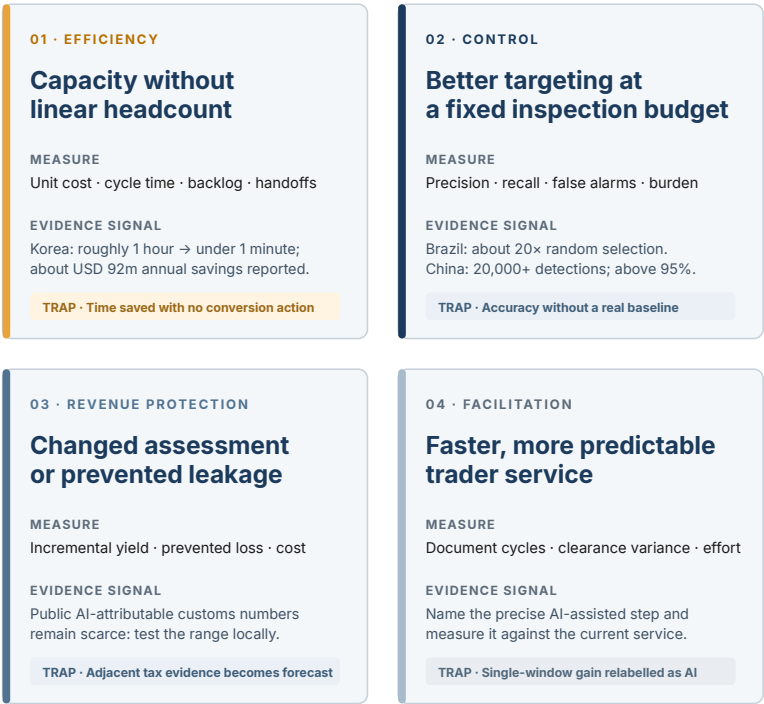
Four value channels

Read each channel through its operating metric, evidence strength, and most common attribution error.

Four Value Channels

EXECUTIVE VALUE LENS

Value appears only when an output changes an operating action.



THE VALUE BRIDGE



No conversion owner = no established business value.

Evidence labels travel with the number: verified · adjacent · practice-reported · directional.

Figure 3.2 - Four Value Channels

The value bridge

The system output is not the benefit. A summary creates value only when it shortens a named officer step. A risk score creates value only when the inspection plan changes and the review burden remains manageable. An agent creates value only when completed work changes capacity, service, revenue, or control.

Build the economic record in this order:

1. baseline volume, time, quality, exceptions, and cost;
2. eligible share and realistic adoption ramp;
3. measured time or quality delta by evidence grade;
4. conversion action—what management actually changes;
5. permanent model, cloud, integration, data, security, oversight, and change costs;
6. downside provision and fallback;
7. sensitivity across volume, unit price, adoption, and exception rate.

Avoided headcount growth is often more credible than layoffs. Faster work has no cash value without a named benefit owner and a decision that changes staffing, service, or risk coverage.

Readiness routes; it does not rank

Assess readiness per task and rung across leadership, legal/sovereignty, data/operations, people, and governance. Require an owner, approved corpus/data flow, deployment, identity/logs, evaluation set, supervisor, mandate, and gate.

An approved read-only Rung 1/L1 service may proceed while a trader-facing write agent does not. The weakest binding dimension sets the task ceiling, not the whole portfolio.

The sovereignty decision

Choose on-premises, recognised government/sovereign cloud, or a hybrid by data class and workload; a commercial region is not accreditation. Counsel and the security authority verify provider, service, subprocessors, data path, terms, and purpose.

Inventory consumer tools and personal API keys; stop sensitive uploads, provide a sanctioned sandbox, and migrate, contain, or retire each workflow by a date. A prohibition without a transition route invites shadow IT.

Core Tool B: Minimum metric spine

Every metric needs a baseline, target range, stop threshold, owner, evidence source, and review cadence. Report ranges and relevant strata, not one headline average.

View	Minimum decision metrics
Operating value	Completed cases · cycle/backlog/handoffs · unit cost · conversion action and realised range
Quality	Complete-case success · evidence/citation validity · abstention/override · repeat stability · burden by relevant strata
Authority and people	Rung/agency · denied tools/scope violations · queue length/age · review minutes · calibration/stop use
Security and sovereignty	Hostile-content/tool test · incidents · data egress · approved region/subprocessor drift
Reliability and change	Retry/loop/tool/state error · release/material change · rollback/demotion result
Evidence, exit, gate	Trajectory/evaluation/export coverage · fallback/exit rehearsal · residual-risk owner · promote/hold/demote/stop · next review

Board rule: no metric is decision-grade unless the leadership team knows what action changes when it crosses the target or stop threshold.

Buying Without Surrendering Judgment

4

Build, buy, partner

The useful boundary is not source code versus subscription. It is institutional judgment versus commodity capability.

DECIDE	EVIDENCE	STOP
Choose what to buy, where to partner, and what judgment must remain inside.	Representative cases, decision rights, evidence export, comparable cost, and exit.	The supplier cannot demonstrate lawful deployment, trajectory export, or fallback.

Buy OCR, translation, infrastructure, identity, monitoring, generic workflow, and other mature layers when the market can supply them under lawful terms. **Partner** where the problem is strategic but engineering and model capability move too quickly to carry alone. **Build and retain** the administration’s problem definition, data meaning,

risk logic, evaluation, governance, operating ownership, and ability to exit.

Korea demonstrates what long-term internal capability can compound. It does not offer a template that can be bought quickly: its conditions include a decade of investment, national operational data, a stable technical cadre, partnerships, and administrative continuity. The transferable principle is not “build everything.” It is: buy the tools, partner on the build, own the judgment.

The RFP that cannot buy the wrong thing

Specify the decision and operating boundary, not a fashionable solution. For every task require the bidder to answer:

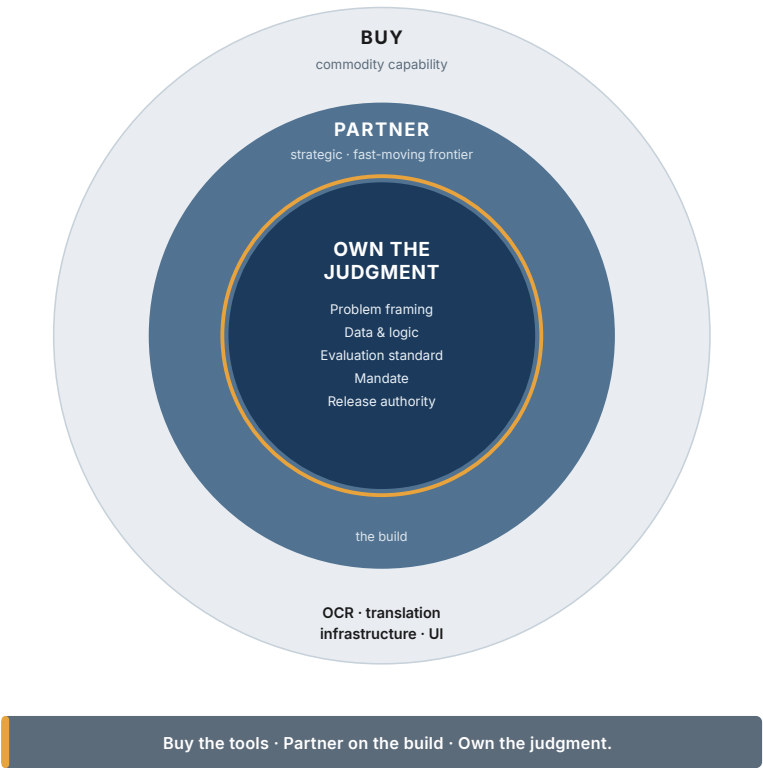
- outcome, baseline, users, and exceptions;
- autonomy rung and why not lower;
- agency profile—tools, data, identities, permissions, writes;
- draft mandate, exclusions, human route, and signer;
- lawful deployment, region, subprocessors, retention, and training use;
- evaluation protocol, cost record, trajectory export, incident path, demotion, fallback, and exit.

The buyer owns the evaluation. A bidder may help design it but does not choose all test cases, define success after seeing the result, or keep the only copy of the evidence.

The Judgment Boundary

What to buy, what to partner on, what to own

Source the commodity, partner on the fast-moving frontier — but keep the judgment inside the administration.



Amber ring marks where decision authority stays — inside the administration.

Figure 4.1 — The Judgment Boundary

Use twenty cases correctly

Give the vendor twenty representative, unseen cases from the real workflow and observe the entire path: ingestion, evidence, intermediate choices, latency, retries, failure, human effort, and unit cost.

Include ordinary messiness—languages, duplicates, missing pages, poor scans—not only adversarial curiosities.

This is qualitative pre-qualification. Twenty cases can reveal integration theatre, hidden manual work, missing citations, and a brittle demo. They cannot estimate a production error rate or establish safety. Formal acceptance uses a stratified, repeated, buyer-controlled evaluation, sealed holdout, hostile cases, baseline comparison, and pre-committed thresholds.

Contract the operating system

The critical clauses cover more than model performance:

Rights and evidence. The administration retains lawful use of its source and transformed data, annotations, feedback, evaluation sets, outputs, performance records, configurations, mappings, and documentation needed to audit and migrate. Vendor background IP can remain vendor IP without trapping the public record inside it.

Change. Model, prompt/policy, corpus, tool, identity, connector, region, and critical dependency changes are classified, tested, recorded, and re-authorised when material. “Continuous improvement” is not permission for silent authority expansion.

Operations. Service levels cover evidence freshness, exception queues, trajectory availability, containment, fallback, and change—not only platform uptime.

Exit. Require readable exports, documented interfaces and mappings, transition assistance, termination capacity, credential revocation, and a rehearsed fallback. Ask annually: if the contract ended in twelve months, what exactly could the administration take and operate?

Commercial transparency. Price inference, storage, integration, managed operations, human review, data work, security testing, change, and exit by task. A cheap model with an expensive exception queue is not a cheap service.

The evidence room

From month one, keep an administration-controlled record of mandate cards, architecture/data flows, component inventory, test sets/results, releases, trajectories, security findings, incidents, training, interfaces, runbooks, and contract evidence. Export it before final acceptance. This converts lock-in from a general fear into testable missing artifacts.

BOX 4.A Local-law adaptation required

The decision-rights clause below is a control template, not procurement or contract advice. Local counsel adapts IP, audit, liability, records, security, and remedy clauses. When direct ownership or source-code access is unavailable or disproportionate, use a lawful functional substitute— tested escrow, readable exports, interfaces, independent assurance, transition assistance, and fallback.

Core Tool D: Decision-rights clause

BOX 4.B Model clause

For **[named task]**, the Contracting Authority retains sole decision rights over the mandate; autonomy and agency; approved data, tools, identities, regions, and subprocessors; evaluation and thresholds; production; material changes; incidents; demotion or stop; acceptance; renewal; and exit.

The Supplier may operate only inside the signed mandate. No default, update, optimisation, or claimed improvement may expand task, authority, permissions, data use, region, dependency, or affected-person scope.

Before acceptance and each material release, the Supplier provides the component/subprocessor register, data/tool flows, evaluation results, trajectory export, change record, incident evidence, rollback, readable exports, and transition material needed to reproduce and investigate.

Use, payment, elapsed time, or silence does not authorise a material change. Only **[named role]** may approve it after the agreed evidence, tests, and review are recorded.

Schedule fields: owner · baseline · rung/agency ceiling · exclusions · deployment/data path · change triggers · gate date · fallback · exit package.

Acceptance test: deny a tool · stop the workflow · export evidence · restore fallback · change supplier without losing the public record.

Govern the Task, Not the Label

5

A three-layer legal view

Every deployment sits inside three layers. **Binding law** covers data protection, secrecy, records, cybersecurity, administrative procedure, procurement, employment, and increasingly AI-specific duties. **Government policy and assurance** turns principles into hosting approvals, impact assessments, model registers, testing, and purchasing conditions. **Soft law and standards**—WCO, OECD, NIST, ISO/IEC and sector guidance—often becomes operationally hard when it enters a tender, audit, insurer, or ministerial approval.

DECIDE	EVIDENCE	STOP
Decide whether the named task has earned a bounded right to operate.	Applicability record, assurance evidence, human route, security tests, and owner.	Legal status, data path, agency reach, or residual-risk ownership is unresolved.

The EU AI Act remains the most influential reference regime. Border management, migration/asylum, biometrics, and some law-enforcement uses can fall into Annex III high-risk categories. Following the Digital Omnibus adopted on 29 June 2026, the relevant date for standalone high-risk systems in customs is 2 December 2027; embedded-product high-risk rules apply from 2 August 2028. Article 50 transparency solutions have a 2 December 2026 timetable. Application still depends on the exact task, deploying entity, people, purpose, and jurisdiction.

Other regimes do not collapse into an EU checklist. Data localisation, automated-decision rights, high-impact definitions, public-law duties, government cloud rules, and regulator guidance differ across the GCC, MENA, Asia, and trading partners. Build a strong common control base, then perform a named-task legal delta review.

BOX 5.A Portable baseline, not compliance guarantee

Human oversight, explainability, documentation, auditability, residency discipline, and standards alignment reduce redesign. They do not certify compliance or make “meaningful human involvement,” “high impact,” or automated decision-making uniform across jurisdictions. Local counsel confirms applicability and required evidence.

From requirement to control

Do not ask legal to approve “the AI programme.” For each task, record: source and applicability · concrete obligation · operating control · evidence that the control works · owner · re-test trigger. A human-oversight duty becomes an interface, named role, escalation, override log, training, and capacity. A transfer restriction becomes an approved

region, subprocessor list, encryption boundary, contract right, and test.

Maintain a quarterly regulatory watch list with official source, status, affected mandate cards, owner, next review, probability over 12–24 months, no-regret preparation, and irreversible commitment to avoid. A proposal is not enacted law; a watch list must not turn it into one.

Autonomy × agency

Oversight follows both the decision rung and technical reach. High agency with broad write tools may require stronger identity, approval, and monitoring even at L2. High autonomy with narrow, reversible, read-only reach may be manageable under sampling and anomaly review. Route by regulatory impact, affected-party proximity, reversibility, data sensitivity, agency, and human exception capacity.

Three human patterns remain useful:

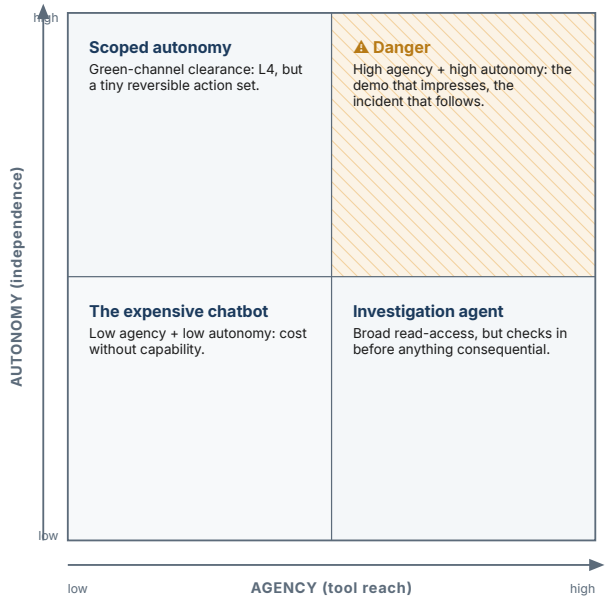
- **human in the loop:** confirms each consequential action;
- **human on the loop:** supervises bounded execution and can intervene;
- **human in command:** sets mandates, resources, audit, promotion, demotion, and retirement.

“Human oversight” is not satisfied by a person theoretically able to watch an impossible queue. Measure queue size, exception age, comprehension, override quality, fatigue, and authority to stop.

Autonomy × Agency

Two independent dials, set on purpose

Autonomy is how independently a system acts; agency is how much it can touch. Grant each deliberately.



Grant capability or independence deliberately — never both by default.

Figure 5.1 — Autonomy × Agency

The operational assurance record

An operational assurance record is the concise argument that a named task may operate at a specified rung with stated residual risk. It links the mandate, data flow, evaluation by relevant strata, hostile-input and tool-misuse tests, trajectory samples, human calibration, inci-

dent/fallback exercise, owner, and dated sign-off. Review it before pilot, autonomy expansion, material change, renewal, and after an incident.

In this book, an *operational assurance record* is an internal governance artifact. It is not a regulatory filing, conformity assessment, certification, or claim of equivalence with aviation, medical-device, nuclear, or other formal safety regimes. Use the locally required name while preserving the evidence logic.

Rights, fairness, and contestability

Where a workflow influences a trader, traveller, or employee, define what is communicated, what evidence can be seen, how a meaningful human review is requested, how the original output is isolated, who can correct the record, and how recurrent errors feed change control. Explanation is not a paragraph generated by the same model. It is the usable connection between source evidence, policy, outcome, and review.

Fairness testing follows the actual service: language, document quality, commodity, route, trader segment where lawful, exception type, and operational burden. Monitor both model error and who absorbs the delay.

Agentic security

Treat trader documents, email, web content, and external-agent messages as hostile. Separate **content** from **control**: untrusted text may inform evidence, but cannot become instruction or expand permission.

Require distinct identities, least agency, enforced tool allowlists, argument and transaction limits, externalised secrets, immutable trajectories, anomaly detection, signed updates, component provenance, tested fallback, and recurring hostile-document tests. “Read only” is a claim until scopes prove it.

Core Tool C: Regulatory applicability checklist

Complete one per task before procurement, production, or an autonomy increase. Record the official source, reviewer, and unresolved owner.

Check	Question that must be answered
Task, place, role	Workflow, jurisdiction, public role, mandate, and decision?
People, effect, risk	Affected people; review, fairness, biometric, migration, law-enforcement, profiling, or high-risk trigger?
Data, records, rights	Secrecy/privacy; purpose, retention, transfer, records, explanation, and contestability?
Sovereignty, security	Hosting/localisation; identity, cyber, supplier, procurement, delegation, records, and audit duties?
AI-specific duties	Transparency, impact/conformity, testing, literacy, monitoring/reporting, and dates?
Evidence and change	Control/evidence owner; release, task, incident, or date review trigger?

Output: applicability record · prohibited/high-risk finding · control owner · counsel question · next review · approve, condition, or stop.

Pilot, Production, and People

6

Pilot the operating system

A useful pilot tests more than whether a model can produce an answer. It tests whether the administration can supply lawful evidence, operate the human gate, observe behaviour, absorb exceptions, control cost, respond to failure, and decide against its own enthusiasm.

DECIDE	EVIDENCE	STOP
Promote, hold, demote, or stop one task at one stage gate.	Complete-case quality, queue capacity, hostile tests, economics, export, and fallback.	Any pre-committed quality, security, authority, or human-capacity threshold fails.

Use four stages.

Design and offline evaluation. Map the current workflow and baseline. Define the task, mandate, autonomy ceiling, agency profile, affected

Piloting in Stages

Taking an agent from trial to production safely

Each stage runs on live data; a human-owned gate, with criteria set in advance, controls the move to the next.



Tones borrow from the Autonomy Ladder (Ch. 2). Amber marks the human-owned gate between stages.

Figure 6.1 — Piloting in Stages

people, failure modes, test strata, thresholds, and fallback. Evaluate on buyer-controlled data with repeated trials and a failure taxonomy.

Shadow mode. Run against real or authorised representative traffic without affecting outcomes. Compare with the current path. Measure consistency, evidence quality, abstention, overrides, latency, unit cost, security, and the exception queue.

Controlled live pilot. Allow a narrow user group, limited volume, and reversible actions. Staff the human route as production work. Exercise freeze, evidence preservation, containment, communication, and fallback.

Limited production. Approve only the tested mandate. Canary material changes, retain a holdout, sample trajectories, and publish a gate memo: promote, hold/tune, demote, or stop.

Reliability compounds across steps

Single-shot accuracy disguises workflow risk. If five dependent steps each succeed 95 percent of the time, the probability of an error-free chain is about 77 percent before correlations and adversarial conditions. Measure complete-case success, state transitions, retries, loops, tool errors, incorrect writes, and escalation—not only the final text.

Repeat identical and equivalent cases. Generative systems are stochastic; a result that passes once may fail under another run, model release, prompt, corpus, or tool state. Pre-commit the number of trials and acceptance rule.

The production control room

Join six views around the task: volume/service · quality/overrides · tools and trajectories · security/incidents · unit economics · mandate and owner. The meeting ends in a decision, not a dashboard: continue, tune, hold, demote, expand, compete, or retire.

Govern four change classes at minimum—model, prompt/policy, corpus, and tool list—and extend the register to identities, connectors, region, workflow, and critical dependencies. Every material change carries owner, test, canary/shadow path, communication, evidence-room update, rollback, and re-authorisation decision.

The second use case is the platform test. It should reuse identity, logging, evaluation, security, evidence, deployment, procurement, and control-room components. If almost nothing transfers, the first project built an implementation, not an institutional capability.

The workforce does not disappear

The officer role shifts with the ladder: Operator, Pilot, Supervisor, Governor. Higher autonomy usually reduces routine touches but increases exception complexity, calibration, sampling, incident work, and the need for people able to challenge both system and policy.

Use a hub-and-spoke model: a central AI/data team owns shared engineering, evaluation, security patterns, and standards; officer-analysts in operational units own domain meaning, exceptions, feedback, adoption, and service outcomes. Pair them. A central team without operations builds generic tools; isolated units duplicate platforms and controls.

Hub and Spoke

Central expertise, embedded in the operational units

A central hub builds and runs the tools; each operational unit holds an analyst paired with a domain expert.

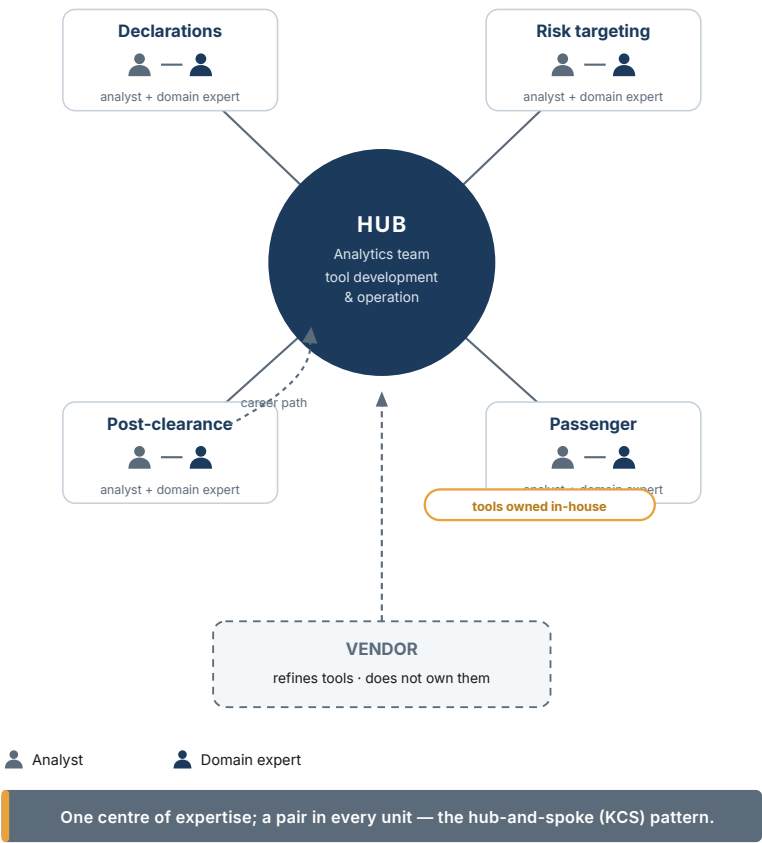


Figure 6.2 — Hub and Spoke

Train by role. Users learn evidence limits, uncertainty, secure handling, and correction. Supervisors learn queue management, sampling, trajectory review, demotion, and incident escalation. Owners

learn mandate and residual-risk decisions. Procurement/legal learn agent definition, agency, evaluation, change, audit, and exit. Engineers learn hostile content, identity, observability, and reproducible release evidence.

Capacity is a control

Set the authorised task volume against staffed review capacity. A system that creates exceptions faster than qualified people can resolve them has exceeded its safe operating envelope even if model metrics remain stable. Define maximum queue length and exception age, surge arrangements, prioritisation, and automatic demotion or fallback.

The “silicon workforce” metaphor is useful only operationally: agents need assignment, permissions, supervision, measurement, change control, and retirement. It does not make software an employee, legal person, public authority, or accountable actor.

Scale gate

Scale only when the local record answers: value captured, not merely time saved · quality stable across relevant strata · human route staffed · security exercises passed · lawful deployment unchanged · unit economics inside range · evidence export works · fallback rehearsed · owner signs residual risk. A successful demo is not on this list.

Core Tool E: Pilot gate template

Complete before every stage change; use one task and proposed rung.

Gate field	Decision record
Task, authority, baseline	Task/owner: ____ · rung/agency: ____ · baseline/holdout/strata: ____
Value and quality	Completed cases/conversion/unit cost: ____ · success/abstention/override/burden: ____
Human route	Reviewer: ____ · maximum queue/age: ____ · calibration/stop authority: ____
Security, sovereignty, change	Hostile-content/tool test: ____ · approved path: yes/no · release/rollback: ____
Evidence and exit	Trajectory/export: ____ · incident/fallback/exit rehearsal: ____

Gate decision: pass / conditional / fail · residual-risk owner: ____ · promote / hold / demote / stop · authorised volume/duration: ____ · automatic stop trigger: ____ · next gate: ____.

Sign-off record

Approval function	Name	Signature	Date
Operations	_____	_____	_____
Legal / privacy	_____	_____	_____
Security	_____	_____	_____
Data / IT	_____	_____	_____
Accountable executive	_____	_____	_____

Cases as Transferable Evidence

7

Read the condition before the result

Cases are useful only when they answer what must be true locally. For each, record the source workflow, technology precisely identified, deployment model, measured outcome, evidence grade, elapsed institutional investment, data conditions, workforce, governance, and what is not public. Then write the local difference. The gap becomes a hypothesis and pilot design—not a reason to copy the headline number.

DECIDE	EVIDENCE	STOP
Choose which external lesson is transferable to the local workflow.	Named technology, evidence grade, denominator, elapsed investment, and local delta.	A headline percentage is imported without its conditions or a local test.

Case Studies at a Glance

FIGURE 15.1

Seven administrations, one reading protocol

Each case on five fixed dimensions — with the technology named precisely, never just “AI,” and its evidence grade.



Read cases consistently: precise technology, honestly graded value.

Figure 7.1 — Cases at a Glance

Korea — sequencing and compounding

Korea Customs Service publicly describes eleven production AI models, national analytics infrastructure, X-ray/manifest integration, and a later natural-language layer. It reports high-risk cargo analysis falling from roughly one hour to under a minute, e-commerce volume rising about 2.85 times without matching workforce growth, and estimated annual savings near USD 92 million.

Transfer: build data access, platform, officer/analyst collaboration, and operating feedback before the generative layer; retain internal capability; communicate outcomes, not launches.

Boundary: a decade of cumulative investment, national operational data, a stable specialist cadre, research partnerships, and institutional continuity cannot be purchased in one tender. The public record does not fully disclose the governance artifacts behind every automated decision. No Korea number enters a local business case until the transfer sheet is complete.

China — image analysis at scale

China GACC reports an AI Image Analysis System across hundreds of non-intrusive inspection devices, algorithm selection, and more than 20,000 detections since 2022 with recognition accuracy above 95 percent.

Transfer: pair sensor/image data with the manifest and workflow; measure human-resource effect and false alarms; treat algorithm routing as part of the governed system.

Boundary: national scale, infrastructure, data, and legal context differ. Administration-reported outcomes do not establish a benefit range for another country.

Brazil — explainability as an operating feature

SISAM combines machine learning with human-readable reasons for targeting and has operated since 2014. Public sources report inspections around twenty times more effective than random selection.

Transfer: explanations are part of workflow acceptance and officer learning, not a compliance paragraph added after the model; compare with a real baseline.

Boundary: long-running ML targeting is evidence for disciplined analytics and explanation, not automatic proof for a GenAI agent or higher autonomy.

India, Netherlands, Uruguay, and the United States

India shows the value of codifying messy supplier identity before advanced analytics. The Netherlands shows computer vision and cooperative research routes. Uruguay shows that a smaller administration can build a focused customs analytics capability without copying the largest administrations. US CBP examples show both operational potential and why a single seizure or testimony anecdote is not a programme metric.

Across these cases, the repeatable assets are narrower than the headlines: entity resolution, data stewardship, bounded workflow choice, officer integration, evaluation, and institutional learning.

Adjacent and practice-reported evidence

Financial-crime detection offers a strong method analogy: rare adversarial events, brittle rules, network relationships, and expensive false positives. Enterprise service operations offer evidence about governed autonomy and the Operator-to-Supervisor shift. Consultancy cases reveal how organisations redesign procurement, compliance review, and assurance.

ADJACENT EVIDENCE — HYPOTHESIS, NOT FORECAST.

Borrow the network method, exception design, and failure mode. Do not transfer the reported percentage into customs.

PRACTICE-REPORTED. Public consultant cases can inform workflow and commercial questions. Attribute the source, retain the client and method limitations, and require a local baseline and pilot.

The cautionary museum

Recognise five exhibits:

- a launch accuracy softened by the speaker in the same sentence;
- a seizure total attributed vaguely to “AI and big data”;
- an “up to” percentage with no primary evaluation;
- a real single-window or process gain relabelled as AI;
- a pilot that passes curated cases but never secures data, ownership, security approval, operating budget, or a staffed exception route.

The fifth is especially important. A proof of concept can be technically successful and institutionally negative. If there is no conversion path, the programme tested the model before it tested the administration.

The transfer sheet

For every external case, complete: source and date · named workflow · technology · evidence grade · metric denominator · starting baseline · elapsed investment · data and legal conditions · workforce · governance · unknowns · local differences · local hypothesis · pilot threshold · expiry date. An incomplete sheet keeps the case out of the financial model.

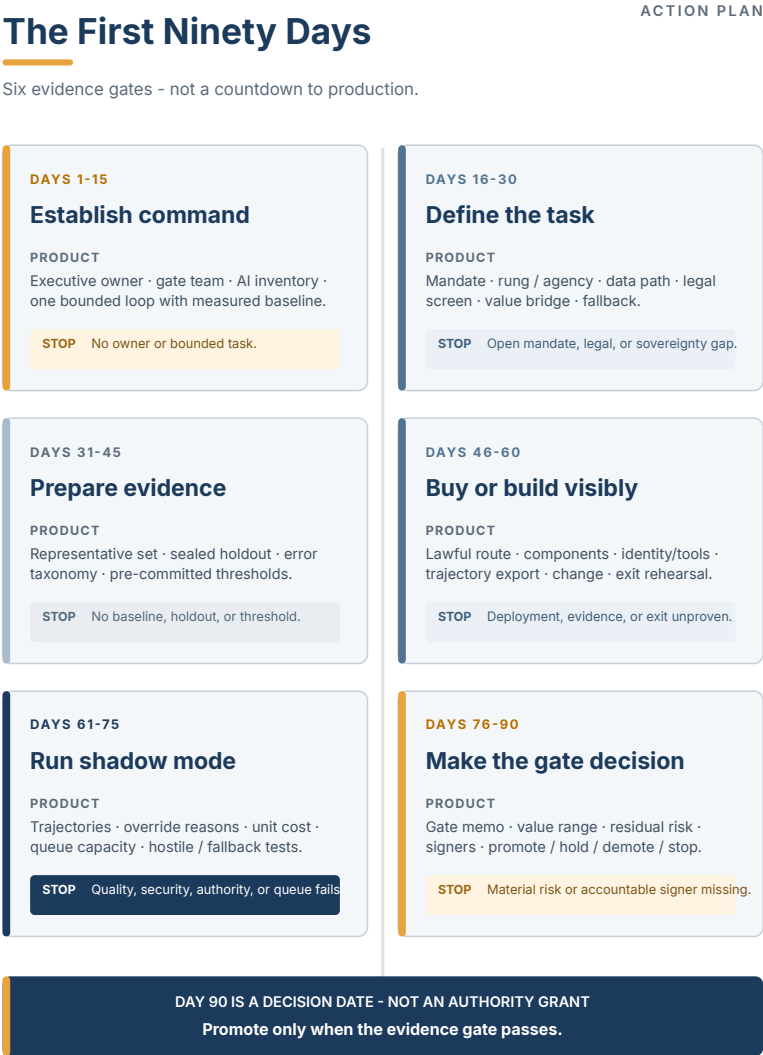
What to Do on Monday

The first ninety days

The first quarter should create decision-grade evidence and shared infrastructure, not a large platform commitment.

DECIDE	EVIDENCE	STOP
Produce a day-90 gate decision, not a technology launch.	Owner, task baseline, mandate, evaluation, lawful route, shadow record, and signers.	Calendar progress overrides an unresolved material risk or missing owner.

The plan is built as six fifteen-day evidence gates. Each stage produces a decision artifact and names the condition that prevents progression.



ESCALATION PATH

Named owner → executive sponsor / gate chair → hold, narrow, simplify, or stop.

A stopped pilot that prevents an unsuitable production system is a successful outcome.

Figure 8.1 - The First Ninety Days

The board dashboard

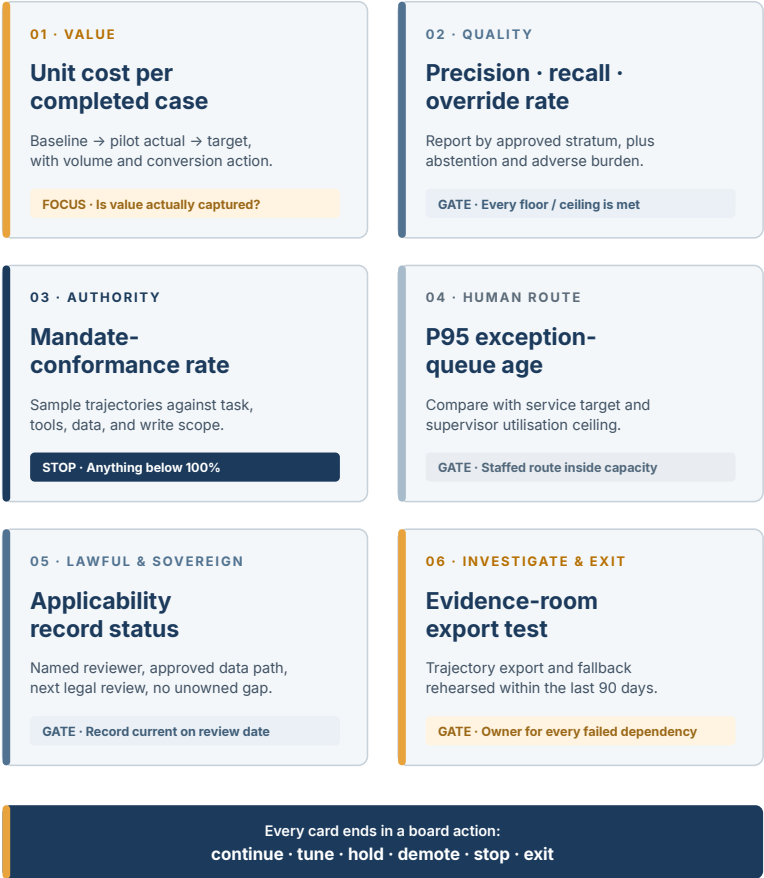
Keep it decision-oriented. Each card shows one primary signal, the evidence needed to interpret it, and the threshold that changes the board decision.

DECIDE	EVIDENCE	STOP
Read one primary operating signal per card.	Baseline, current result, target or stop threshold, owner, and source.	Do not close a review without a named continue, tune, hold, demote, stop, or exit action.

The Board Dashboard

EXECUTIVE BRIEF

Six questions. Six measurable signals. One decision at every review.
Show the baseline, current result, target or stop threshold, owner, and action - not a status colour alone.



MINIMUM RECORD
Baseline · current · target / stop · evidence source · owner · next review · action

A dashboard is decision infrastructure, not a collection of green status lights.

Figure 8.2 - The Board Dashboard

The 12–24 month watch

Maintain one row per signal: capability, cost, sovereignty, trader-side agents, standards, and law. Each row records probability, earliest plausible decision date, official source, no-regret preparation, irreversible commitment to avoid, readiness, owner, and next review. These are the complete minimum fields for this brief. Do not buy against a forecast; build options that remain useful if it is wrong.

Five operating rules

01 Govern one real task

Give it a baseline, accountable owner, and an accurate pipeline-or-agent description.

02 Record both dials

Limit autonomy one rung at a time and agency to the reach actually needed.

03 Keep sovereignty and evidence

The administration owns the approved path, evaluation, and exportable evidence room.

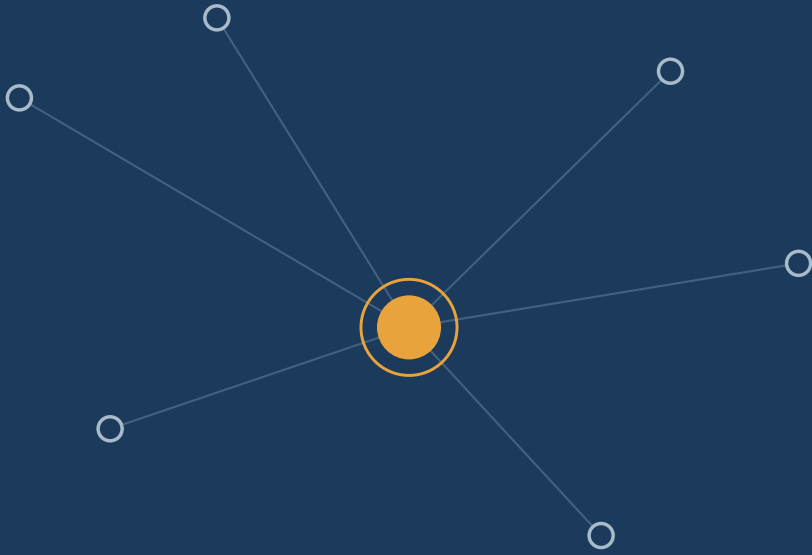
04 Earn the right to operate

Carry a mandate, assurance record, viable human route, demotion trigger, and tested fallback.

05 Treat change and renewal as decisions

Material changes re-enter the gate; renewal may continue, compete, migrate, demote, or retire.

THE FINAL BOARD QUESTION. *The mandate of customs remains collect revenue, protect the border, and facilitate trade. Agentic AI changes the operating model, not institutional responsibility. Per task, ask: what may it decide; what can it technically do; who signed; what evidence earned that authority; and when will we take it back? A leadership team that can answer is running an AI programme, not following a vendor narrative.*



A DECISION BOOK FOR BORDER LEADERS

AI Agents at the Border is a practical guide for leaders who must decide what may be delegated — and what must remain human.

A field guide to buying, governing and operating AI that can act in trade and customs environments.

**DECISIONS MAY TRANSFER WITHIN A SIGNED MANDATE.
ACCOUNTABILITY DOES NOT.**

MAKSYM NECHEPURENKO

Data Science Manager, Agility · UAE

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